Yes, you can write userspace drivers (known as **translators** or userspace server processes) for GNU/Hurd using C++, though standard practice and official libraries lean heavily on C. [1]

How Hurd Translators Work

- **User-Space Nature:** In the GNU/Hurd architecture built on top of the Mach microkernel, device drivers and filesystem handlers run as normal user-space programs (servers or translators) rather than inside a monolithic kernel. [1, 2]
- **IPC via MIG:** Hurd components communicate via **Mach Inter-Process Communication (IPC)** using stub code generated by **MIG** (Mach Interface Generator). [1, 3]
- **Language Support:** Because translators are standard user-space applications, you can theoretically write them in any language that can handle POSIX and Mach/Hurd RPC interfaces. However, official C bindings and header files (libfshelp, libports, etc.) are written for C. [1, 4]

Challenges with C++ in Hurd Translators

- **MIG and C Bindings:** The core Hurd client/server stubs generated by MIG are designed for C. If you use C++, you will need to wrap or write extern "C" interfaces around the Mach/Hurd message-passing and port-management routines.
- **Lack of C++ Libraries/Ecosystem:** Most sample code, documentation, and template translators in the GNU/Hurd source tree are written in C. Using C++ means you are largely on your own for mapping C++ classes to Hurd's port and hash-table management structures (such as libports or hurd_ihash). [4]
- **Alternative Approaches:** Projects working on Hurd hardware support have historically focused on reusing existing driver stacks—such as running NetBSD drivers inside a user-space framework (like the *rump kernel* approach) or utilizing DDE (Device Driver Environment) layers for Linux drivers—rather than building custom object-oriented C++ drivers from scratch. [5, 6]

Are you planning to build a custom translator for a specific **hardware device** or a **virtual filesystem** on Hurd? Let me know so I can point you toward the right Hurd libraries.

[1] https://www.gnu.org
[2] https://www.youtube.com
[3] https://www.youtube.com
[4] https://git.sceen.net
[5] https://www.phoronix.com
[6] https://www.gnu.org